

Han Gao

Undergraduate Student in Software Engineering

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Profile

Undergraduate student at Xi'an Jiaotong University. Incoming graduate student at Zhejiang University in 2027. Research interests include Multimodal Large Language Models (MLLMs), Hyperspectral Image Change Detection (HSICD), cross-modal retrieval, and Multi-Agent Systems (MAS).

Education

M.Eng. Zhejiang University , Software Engineering	Ningbo, Zhejiang, China Starting 2027
B.E. Xi'an Jiaotong University , Software Engineering • School of Software Engineering.	Xi'an, Shaanxi, China 2023 – 2027

Research Experience

Xi'an Jiaotong University , Undergraduate Researcher Multimodal large language models for hyperspectral image change detection. - Reframed pixel-level change detection as a vision-language question-answering task for semantic interpretation. - Refined ground-truth labels and designed an automated pipeline for region-level captions and question-answer pairs. - Adapted multimodal models with supervised fine-tuning and parameter-efficient fine-tuning, and conducted evaluation and analysis.	Xi'an, Shaanxi, China Jan 2026 – present
Xi'an Jiaotong University , Undergraduate Researcher Structure-aware SAR-RGB cross-modal image retrieval. - Investigated CLIP-style alignment and contrastive learning across heterogeneous sensing modalities. - Implemented strip-convolution modules to capture elongated ship structures. - Completed model training, hyperparameter tuning, ablation studies, and comparative analysis.	Xi'an, Shaanxi, China Sept 2025 – Dec 2025
Provincial Undergraduate Innovation Project , Core Team Member Deep learning for cytological diagnosis of pleomorphic adenoma. Contributed to literature review, technical planning, data preparation, model discussion, and experimental analysis.	Xi'an, Shaanxi, China Sept 2025 – Apr 2026

Publications

HSICD: Multimodal Large Language Model for Hyperspectral Image Change Detection Han Gao, et al. (IEEE Geoscience and Remote Sensing Magazine (under review))	2026
A Unified Algorithmic Framework for Battery Life Prediction Based on CTMC and Monte Carlo Simulation Han Gao, et al. (MEEML 2026)	2026

Selected Projects

Multi-Agent Literature Research Assistant A tool-augmented multi-agent system for paper discovery, analysis, literature synthesis, and research knowledge management. - Designed retrieval, analysis, and generation agents using MCP, A2A, and retrieval-augmented generation concepts. - Integrated Zotero and Obsidian for research knowledge-base synchronization. - Built agent tracing for observable coordination and tool-use workflows.	Mar 2026 – present
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Honors

2025: University Second-Class Scholarship; Outstanding Student Leader
2024: University Third-Class Scholarship; Outstanding Student Leader
MCM/ICM: Honorable Mention

Technical Skills

Programming: Python, C++, Java
Artificial Intelligence: PyTorch, CLIP, LoRA, LLM fine-tuning
Agent Development: MCP, A2A, RAG, tool-augmented agents
English: CET-6 (647), CET-4 (661)